

Vision

The written notes for the vision walkthrough. Week 5 competencies 16 to 19.

Step by step: where the signal travels

2 min

The path of light through the eye

1. Cornea, which does about two thirds of the refraction.
2. Aqueous humor in the anterior chamber.
3. The pupil, the opening in the iris.
4. The lens, which adjusts the focus.
5. The vitreous body in the vitreous chamber.
6. The retina. Light passes through the ganglion cell and bipolar cell layers to reach the rods and cones at the back, against the retinal pigment epithelium (except at the fovea, where the inner layers are pushed aside).

Phototransduction in a rod, step by step

1. Dark: cGMP is high, CNG channels are open, Na⁺ and Ca²⁺ enter while K⁺ leaks out, the rod sits near -40 mV, and it releases glutamate continuously.
2. A photon is absorbed; retinal changes shape and is released from opsin (bleaching).
3. Activated opsin turns on the G protein transducin.
4. Transducin's cascade (through phosphodiesterase) lowers cGMP.
5. CNG channels close; Na⁺ and Ca²⁺ entry stops while K⁺ keeps leaving.
6. The rod hyperpolarizes toward -70 mV, in proportion to the light.
7. Fewer voltage-gated Ca²⁺ channels in the terminal are open, so glutamate release falls.
8. Recovery: retinal moves to the retinal pigment epithelium, reverts to its inactive form, and returns to rejoin opsin. This is slow, which is why dark adaptation takes time.

The path of the signal, from photoreceptor to cortex

1. Photoreceptor (rod or cone): graded potentials; light lowers its glutamate release.
2. Bipolar cell: graded potentials. ON bipolar cells are excited by light (less glutamate), OFF bipolar cells

are inhibited by it. Horizontal cells add lateral inhibition.

3. Ganglion cell: fires the first action potentials; amacrine cells modulate the bipolar-to-ganglion signal. Ganglion cell axons form the optic nerve and leave at the optic disk (blind spot).
4. Optic chiasm: fibers from the inner (nasal) half of each retina cross; fibers from the outer half stay on their own side.
5. Optic tract to the lateral geniculate body of the thalamus.
6. Visual cortex in the occipital lobe. Some fibers go to the midbrain for eye movements and the pupillary reflex.

The pupillary light reflex, step by step

1. Light reaches the retina of one eye.
2. The signal travels in that eye's optic nerve, cranial nerve II.
3. It reaches the thalamus; collateral pathways go to the midbrain.
4. In the midbrain the signal diverges to both sides.
5. Parasympathetic fibers in cranial nerve III contract the pupillary sphincter of each eye, and both pupils constrict (the consensual reflex).

Accommodation for a near object, step by step

1. You look from something far away to something near; its light rays are no longer parallel.
2. Parasympathetic signals make the ciliary muscle contract.
3. The muscle ring gets smaller, releasing tension on the zonules.
4. The elastic lens rounds up, shortening its focal length and bending light more.
5. The near image lands on the retina.

Structures of the eye

1 min

Structure	What it does
Orbit	Bony cavity that protects the eye
Extrinsic eye muscles	Six skeletal muscles that move the eye; cranial nerves III, IV and VI
Lacrimal apparatus	Lacrimal gland secretes tears (stimulated by parasympathetic fibers of cranial nerve VII); the nasolacrimal duct drains them into the nasal cavity
Sclera	Outer connective tissue wall; the cornea is its clear front continuation
Iris and pupil	Pigmented ring and its opening; sphincter (parasympathetic) constricts, radial dilator (sympathetic) dilates
Lens and zonules	Clear elastic disk on inelastic ligaments attached to the ciliary muscle
Aqueous humor	Low-protein fluid in the anterior chamber, secreted by the ciliary epithelium, drained through the canal of Schlemm; nourishes the cornea
Vitreous body	Clear gel filling the vitreous chamber; maintains the shape of the eyeball
Retina	Layers of photoreceptors and neurons; the fovea and macula give the sharpest vision; the optic disk is the blind spot
Retinal pigment epithelium	Dark layer behind the photoreceptors; absorbs stray light, recycles retinal, phagocytizes the old disks at the tips of the outer segments, and helps form a blood-retinal barrier

- Glaucoma: degeneration of the optic nerve; raised intraocular pressure from blocked aqueous humor drainage is a risk factor, though not the only cause. Leading cause of blindness worldwide.

- Macular degeneration: loss of central vision; peripheral vision remains.

Focusing light

1 min

- Refraction: light bends when passing between materials of different density. About two thirds happens at the cornea, one third at the lens.
- A convex lens converges light to a focal point; a concave lens spreads it. The focal length is fixed unless the lens changes shape; a rounder lens has a shorter focal length.
- Objects about 6 meters (20 feet) or more away send parallel rays and focus with a flatter lens.
- Accommodation: the ciliary muscle contracts (parasympathetic), the zonules slacken, the lens rounds. The near point of accommodation is the closest distance in focus.
- Accommodation declines from childhood; by 40 it is about half of what it was at 10, and by 60 many people lose it (presbyopia).

	Far object	Near object (accommodation)
Ciliary muscle	Relaxed	Contracted
Zonules	Taut	Slack
Lens	Flatter, less bending	Rounder, more bending

Condition	Cause	Focus falls	Corrective lens
Myopia (nearsighted)	Eyeball too long or cornea too curved	In front of the retina	Concave
Hyperopia (farsighted)	Eyeball too short or cornea too flat	Behind the retina	Convex
Presbyopia	Stiff lens; accommodation lost	Behind the retina for near objects	Convex reading lens
Astigmatism	Cornea not a perfect dome	Distorted image	Lens shaped to the uneven curve

Pupillary light reflex

Light in one eye constricts both pupils (consensual reflex). If light in the left eye constricts the right pupil

but not the left, the efferent path to the left eye is broken. If it constricts neither while light in the right eye constricts both, the afferent path from the left eye is broken.

Depth of field

A narrower pupil lets a narrower beam in, keeping more of the scene in focus.

Photoreceptors and phototransduction

1 min

	Rods	Cones
Light needed	Dim; night vision	Bright; daytime vision
Color	No (black and white)	Yes: red, green and blue cones
Where	Most of the retina; far outnumber cones	Only cones in the fovea
Pigment	Rhodopsin	Three pigments related to rhodopsin
Convergence	High: sensitive, low acuity	Low at the fovea, as little as one to one: high acuity

- Each photoreceptor has an outer segment with pigment disks, an inner segment with the nucleus and organelles, and a synaptic terminal releasing glutamate onto bipolar cells.
- Rhodopsin = opsin + retinal (from vitamin A). Bleaching releases retinal; its slow recycling explains slow dark adaptation.
- Color comes from comparing the three cone types; color blindness is an inherited defect in one or more cone types, red-green being best known.
- Visible light runs from about 400 nm (violet) to 700 nm (red).

Processing in the retina and beyond 1 min

- Five kinds of retinal neuron: photoreceptors, bipolar cells, ganglion cells, horizontal cells (lateral inhibition) and amacrine cells (modulate bipolar-to-ganglion signals). Glia include astrocytes and Müller cells.
- Convergence: many photoreceptors onto one bipolar cell, many bipolar cells onto one ganglion cell; minimal at the fovea, greatest at the edges.
- ON bipolar cells have a metabotropic glutamate receptor that hyperpolarizes them, so they are activated by light. OFF bipolar cells have an ionotropic receptor that depolarizes them, so they are excited in the dark.
- Ganglion cell receptive fields have a center and a surround (on-center or off-center). Strong contrast gives a strong response; even light gives a weak one, so the retina reports contrast.
- M ganglion cells: movement. P ganglion cells: form and fine detail. Melanopsin-containing ganglion cells send light information to the suprachiasmatic nucleus for circadian rhythms; most sensitive to blue light.
- The image on the retina is upside down; the brain reverses it.
- Left visual field of both eyes goes to the right visual cortex, and right to left. The binocular zone gives depth perception; the monocular zones are seen in two dimensions.
- The lateral geniculate body and the visual cortex keep a map of the visual field; the cortex sorts information by form, color and movement.

Lesion	Field loss
One optic nerve, before the chiasm	All vision in that eye
Middle of the chiasm (pituitary tumor)	Outer (temporal) half of each eye's field: bitemporal hemianopia, often unnoticed for a long time